

Ghost in the Machine

Detecting the Mathematical Fingerprint of AI-Generated Text
Across Literary Impersonation

Project Report

Kanishk Kulkarni

mkanishkkulkarni@gmail.com

kanishkez.github.io

July 2, 2026

1 Overview

This report presents *Ghost in the Machine*, a project investigating whether AI-generated text leaves behind a detectable “mathematical fingerprint” regardless of the stylistic guise it adopts. A multi-tiered detection system was built to distinguish authentic human authors (Charles Dickens, Jane Austen) from AI language models attempting to mimic their literary style. Beyond raw detection accuracy, the project asks two deeper questions: does an AI detector generalize across different authors, meaning it has learned something universal about machine generation rather than the quirks of one author’s imitation; and how robust is that detector when an adversary deliberately tries to evolve text past it?

The project is organized around three progressively deeper detectors (stylometric, semantic, and transformer-based), two diagnostic experiments (cross-author transfer and feature ablation), an explainability layer (SHAP and Captum), and an adversarial stress test using a genetic algorithm.

2 Dataset

The dataset consists of 720 paragraphs, evenly split across three classes (240 each):

1. **Human** – authentic excerpts from classic literature (Charles Dickens, Jane Austen), sourced from Project Gutenberg.
2. **AI Generic** – standard AI-generated text with no stylistic constraint.
3. **AI Impostor** – AI-generated text explicitly prompted to mimic a specific author’s style.

This three-way split makes it possible to separate two different questions: human vs. AI origin, and generic vs. deliberately-mimicked AI generation.

3 Methodology

The system evaluates text through three independent detection tiers of increasing representational depth, followed by an explainability layer.

3.1 Tier A – Stylometric Random Forest

A Random Forest classifier trained on 37 handcrafted stylometric features: Type-Token Ratio, syntax/dependency depth, readability scores, punctuation frequencies, and function word ratios. The figures below show how several of these features are distributed across the three classes.

3.2 Tier B – Semantic MLP

A multi-layer perceptron trained on dense sentence embeddings (`all-MiniLM-L6-v2`) to capture deep semantic structure that surface-level stylometry may miss.

3.3 Tier C – Transformer (DistilBERT + LoRA)

A DistilBERT model fine-tuned using Low-Rank Adaptation (LoRA) for direct sequence classification, leveraging pretrained contextual representations while keeping fine-tuning lightweight.

3.4 Explainability

Two complementary techniques were used to interpret model decisions:

- **SHAP** – global feature importance for the Tier A Random Forest.

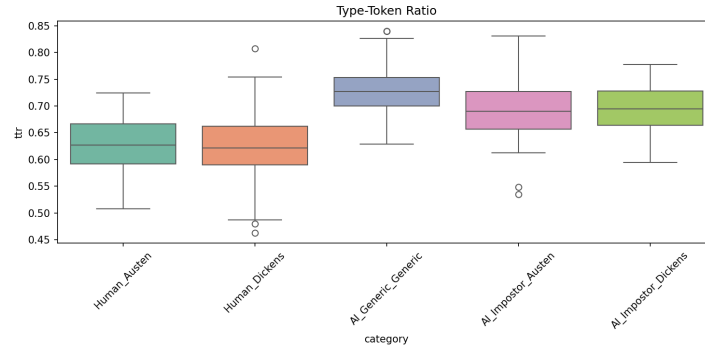


Figure 1: Type-Token Ratio (TTR) distribution across Human, AI Generic, and AI Impostor classes.

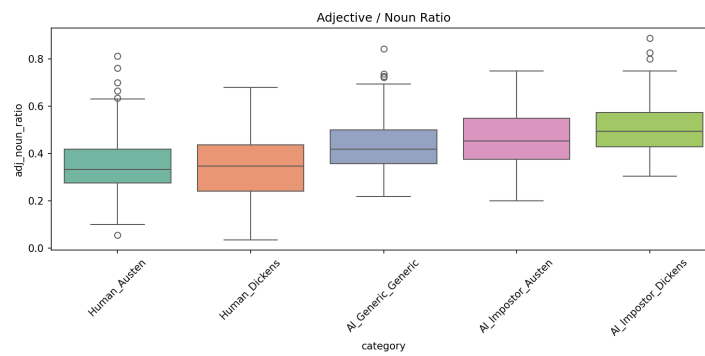


Figure 2: Adjective-to-noun ratio distribution across the three classes.

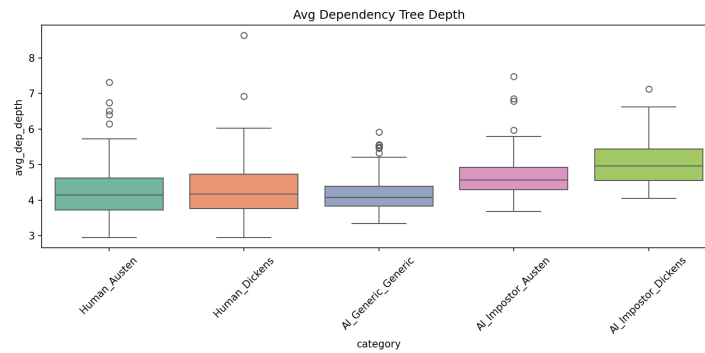


Figure 3: Average dependency parse tree depth distribution across the three classes.

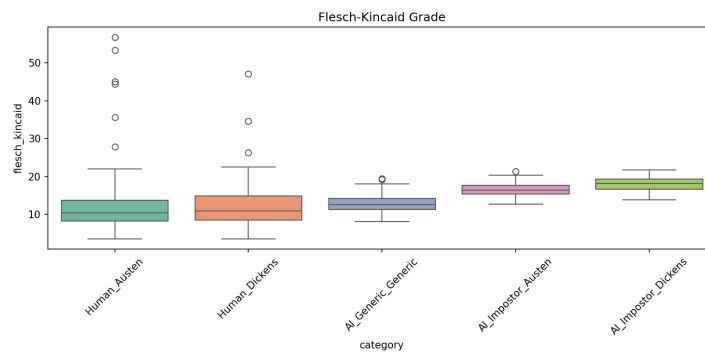


Figure 4: Flesch-Kincaid readability score distribution across the three classes.

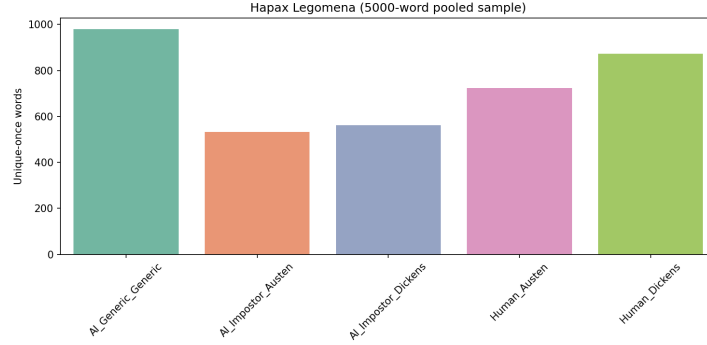


Figure 5: Hapax legomena rate (words occurring exactly once) distribution across the three classes.

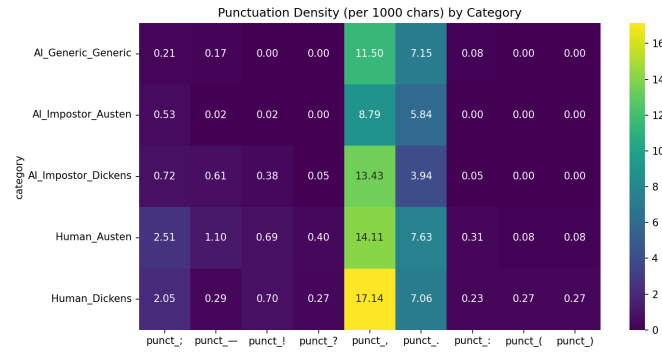


Figure 6: Punctuation frequency usage heatmap across the three classes.

- **Captum Layer Integrated Gradients** – token-level saliency for the Tier C transformer.

4 Results

4.1 Detection Accuracy

Table 1 summarizes overall three-way classification performance across all tiers; Figure 7 visualizes it directly.

Table 1: Detection accuracy and AUC by tier

Tier	Accuracy	AUC
Tier A (Stylometric RF)	97.0%	0.999
Tier B (Semantic MLP)	95.0%	0.995
Tier C (DistilBERT LoRA)	93.0%	0.997

Interestingly, the simplest detector – handcrafted stylometric features fed into a Random Forest – outperforms both the semantic embedding model and the fine-tuned transformer, suggesting that AI-generated text differs from human text in ways that are directly measurable at the surface level, without needing deep contextual representations.

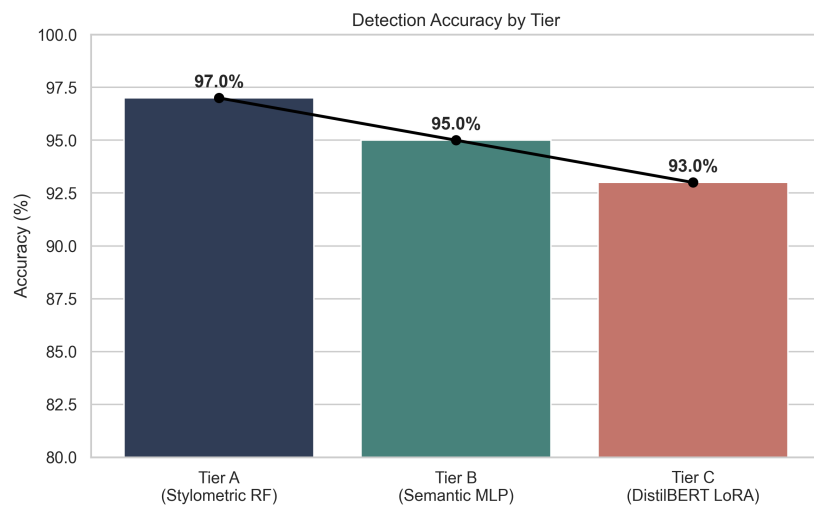


Figure 7: Detection accuracy and AUC by tier.

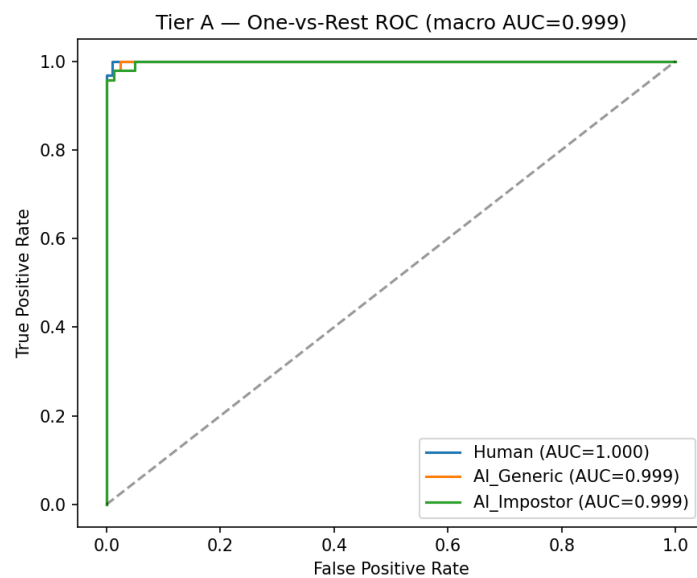


Figure 8: ROC curve for the Tier A stylometric Random Forest classifier.

4.2 Cross-Author Transfer

To test whether the detector learns a universal AI signature rather than memorizing one author’s imitation quirks, each detector was trained on one author and tested on the other, held-out author.

Table 2: Cross-author transfer accuracy

Train → Test	Accuracy
Dickens → Austen	83.8%
Austen → Dickens	95.4%

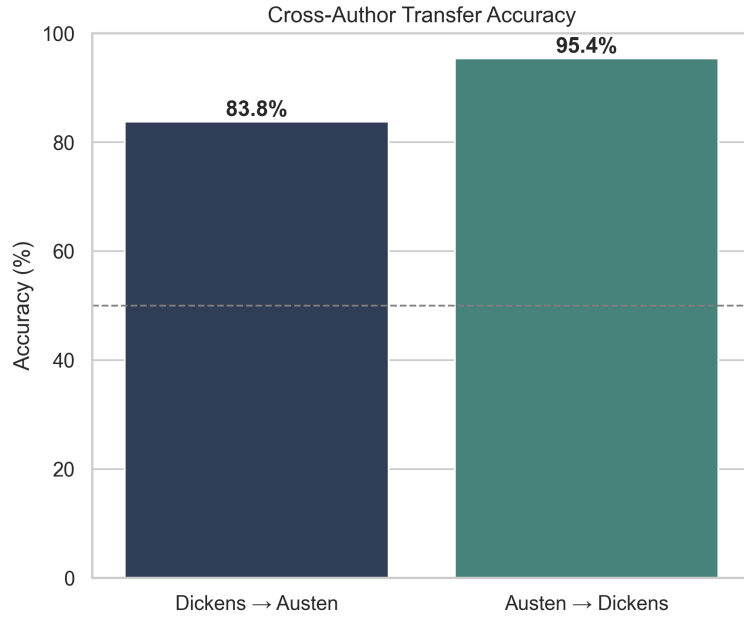


Figure 9: Cross-author transfer accuracy.

Both directions hold up well above chance (33% for a 3-class problem), which is the key finding of the project: the detector generalizes across very different literary styles, supporting the idea of a universal, author-independent AI fingerprint rather than memorized stylistic cues.

4.3 Feature Ablation (Tier A)

The isolated contribution of each stylometric feature group was measured by selectively removing or retaining feature groups.

Table 3: Tier A feature ablation results

Configuration	Accuracy
Full feature set (baseline)	97.0%
Removing Lexical Richness	96.9% (no drop)
Removing Function Words	93.0% (significant drop)
Training on only Function Words	92.2%
Training on only Lexical Richness	62.5%

Removing function words causes by far the largest accuracy drop, and function words alone

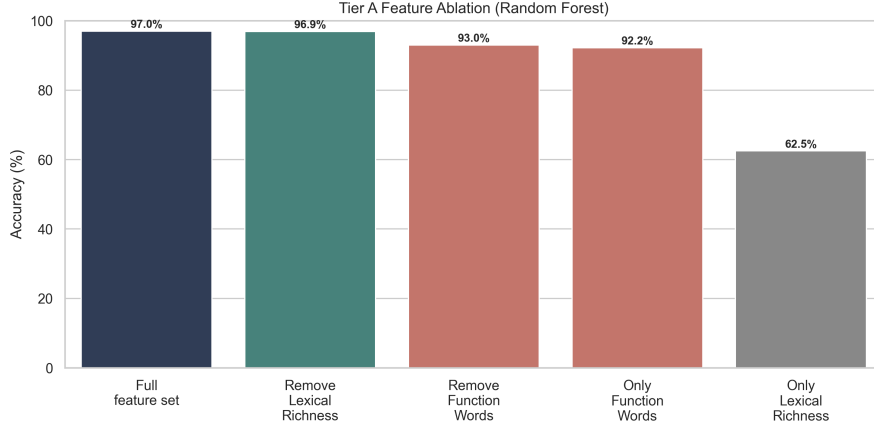


Figure 10: Tier A feature ablation results.

retain almost all of the full model’s discriminative power. Lexical richness alone barely beats chance. This points to function words – the “invisible connective tissue” of language – as the single biggest giveaway of AI-generated text.

4.4 Explainability Results

4.5 The Turing Test: Genetic Algorithm Attack

A genetic algorithm iteratively mutated AI-generated text (archaic word swaps, clause shuffling, rhythm changes) with the goal of maximizing the Tier C detector’s predicted probability that the text is human-written.

Table 4: Genetic algorithm evasion progress

Generation	P(classified as Human)
Generation 0	17.2%
Generation 7	90.3%

The detector was successfully bypassed within eight generations – the mutated text’s probability of being classified as human rose from 17.2% to 90.3%, demonstrating that even the strongest detector in this study is vulnerable to targeted adversarial optimization.

5 Discussion

Taken together, the results support the existence of a detectable, largely style-independent AI fingerprint in generated text. The strongest evidence is twofold: the cross-author transfer results show the detector is not simply memorizing one author’s imitation quirks, and the feature ablation results show that this signal lives mostly in function-word usage rather than vocabulary richness – a low-level, largely unconscious pattern of language use. At the same time, the genetic algorithm attack shows that this fingerprint, while real, is not robust to deliberate, iterative adversarial pressure. Detection and evasion are not symmetric: a detector that is highly accurate against naturally-generated AI text can still be defeated once an adversary is allowed to optimize directly against it.

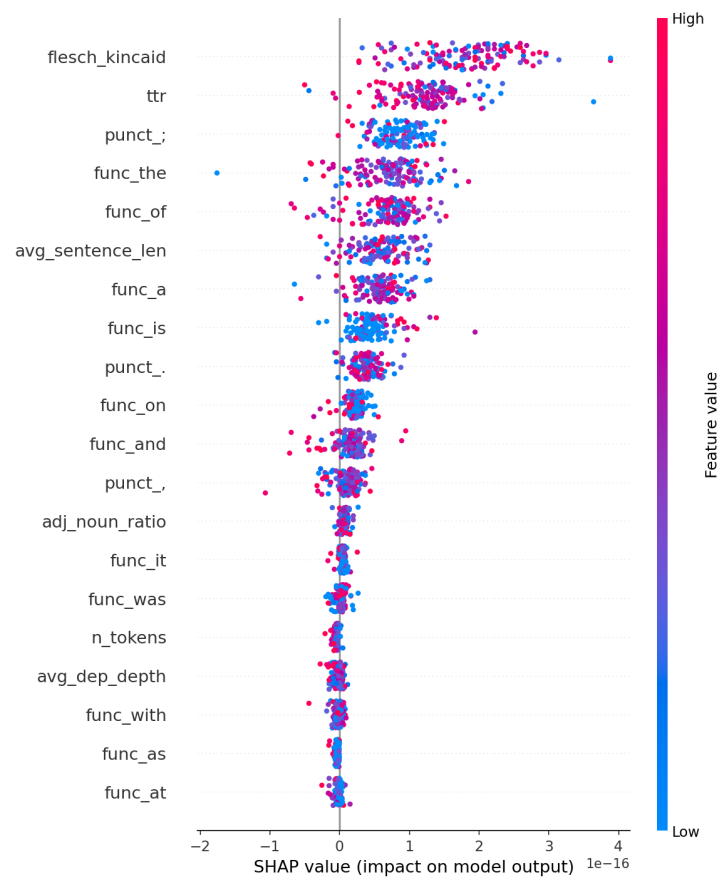


Figure 11: Global SHAP feature importance summary for the Tier A Random Forest.

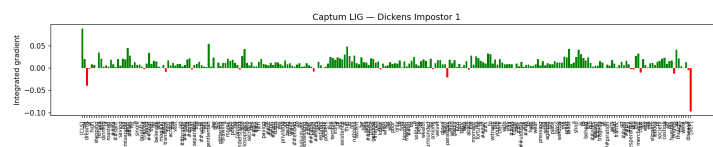


Figure 12: Captum Layer Integrated Gradients token-level saliency for a Dickens-style AI Impostor passage.

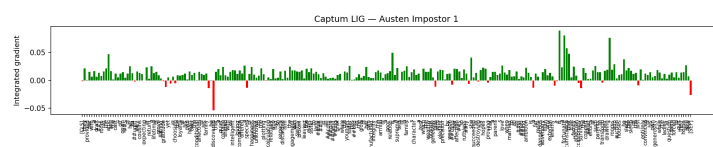


Figure 13: Captum Layer Integrated Gradients token-level saliency for an Austen-style AI Impostor passage.

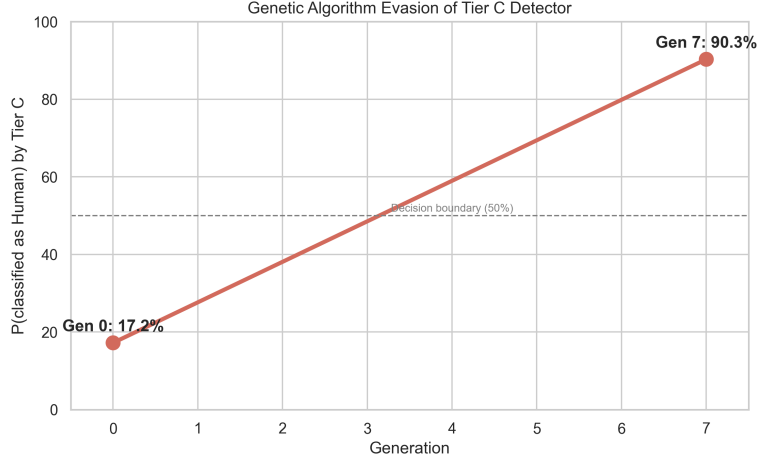


Figure 14: Genetic algorithm evasion of the Tier C detector over generations.

6 Conclusion

This project built a three-tier system for detecting AI-generated text across both generic and author-impersonating generation, reaching up to 97.0% accuracy with a stylometric Random Forest. Cross-author transfer experiments provide evidence for a universal AI fingerprint, and feature ablation traces that fingerprint mainly to function-word usage. SHAP and Captum explainability ground these findings at both the feature and token level. However, a genetic algorithm was able to bypass the strongest detector within eight generations, showing that current detection approaches remain fragile under active adversarial pressure. Natural next steps include ensembling all three tiers, adversarially training against evolutionary attacks, and extending the dataset to a wider range of authors and generator models.